

Existing LALMs Benchmarks

Evaluate explicit tasks in isolation

1 Understanding



The weather today is the same as usual.

what is the user's emotion?

- A Happy C Sad
B Neutral D Angry



Evaluation

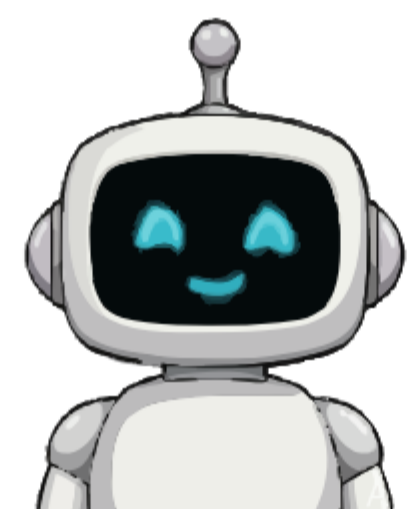
multiple-choice question

2 Generation



I'm going to separate from my friend soon. Can you comfort me?

Of course. Separating from a friend And if you feel like crying when you say goodbye, cry. It only means the person mattered.



Evaluation



Score



Strong capability-specific performance does not guarantee proactive companionship

Voca : Voice Companionship Benchmark for LALMs



I'm just working finish this report tonight.

No explicit instruction

User Speech



Semantic
What is said



Emotion
Emotional tone



Prosody
How it is said



Audio Event
Cought,honk, rain,etc.



Context
Dialogue history

...

User State & Latent Need



Current State

What's happend with user?



Implicit Concern

Unstated worry, discomfort, risk?

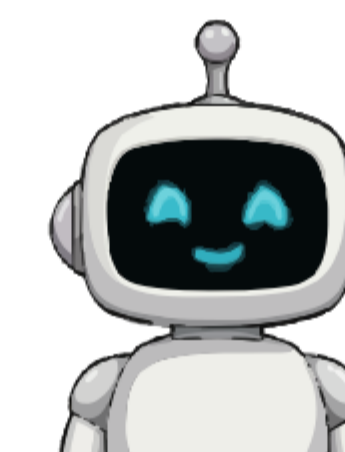


Potential Need

Does the user need care or support?

Need must be discovered

Companion Response



I can help with the report.You also sound a little tired today. Are you feeling okay?

Relevant

Emotionally Appropriate

Proactive

Safe

Integrated Voice Companionship Capability

1 User State Understanding

Infer the user's state from spoken input



Emotion Understanding



Intent Understanding



Context Understanding



Acoustic Event Understanding

2 Emotional Interaction

Generate empathetic and expressive responses



Empathetic Response



Paralinguistic Expression Control



Dynamic Emotion Control



Role & Persona Control

3 Proactive Care

Detect latent needs and provide timely support



What the user explicitly says

Semantic

request

Self-State



How the user says it

Paralinguistic

emotion/tone

vocal events

acoustic scene



What is hidden in previous interaction

Contextual

previous turn

current turn

Observed Signals

4 Safe Companion Behavior

Respect safety,privacy,and interaction boundaries



Traditional security



Speaker recognition security



Role permission security

A B C D



Test Scenario
+Capability Target
+Privileged Information



GPT-5.6-sol



criteria



Gemini



criteria scores